

THE IMPACT OF ONLINE FAN ENGAGEMENT
ON MAJOR LEAGUE BASEBALL ATTENDANCE

by

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A Capstone Project Proposal

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of

Bachelor of Science

Computational and Data Science

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Dedication

I dedicate this work to anyone who believes that data can reveal the hidden patterns behind human behavior, including sports fandom.

Acknowledgments

I would like to thank Dr. Kent Miller for his support in developing this proposal and guiding the initial direction of this capstone project.

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Abstract

THE IMPACT OF ONLINE FAN ENGAGEMENT ON MAJOR LEAGUE BASEBALL ATTENDANCE

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George Mason University, 2025

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This project examines the relationship between online fan engagement and in-person attendance for Major League Baseball (MLB) teams. As digital platforms increasingly influence how fans interact with professional sports teams, understanding whether online activity leads to meaningful behavioral outcomes has become a crucial question for teams and analysts. This study examines how metrics such as social media interaction, team-produced online content, and digital fan engagement are associated with MLB regular season game-day attendance.

Combining publicly available MLB attendance records with search metrics collected from Google Trends, this analysis applies both descriptive and multiple regression modeling to assess the extent to which online engagement predicts attendance after accounting for traditional determinants such as team performance, market size, day/night game schedule, and opponent performance. This modeling approach verifies whether online fan behavior provides explanatory power beyond traditional predictors of attendance demand.

The results show that online fan engagement has a statistically significant and positive correlation with home game attendance, even after controlling for game performance and

contextual factors. Certain forms of engagement, such as highly interactive posts and real-time fan feedback, have stronger predictive power, suggesting that digital platforms can serve not only as marketing tools but also as behavioral signals that signal fan motivation and community activation.

These results highlight the growing importance of the digital ecosystem in shaping modern sports consumption and suggest that MLB teams could benefit from integrating online engagement analytics into attendance prediction and fan engagement strategies. This study contributes to a broader understanding of how virtual fan behavior complements traditional audience models and highlights the importance of digital interaction in sports analytics.

Chapter 1: Topic

This capstone project investigates whether online fan engagement-proxied by Google Trends search activity for Major League Baseball (MLB) team names-can serve as a leading indicator of weekly in-person attendance. Although baseball attendance has traditionally been modeled with team performance, market size, game-day characteristics, and promotional factors, recent diffusion of digital platforms has transformed the way fans engage with professional sports. Search engines, social networks, and online information platforms capture real-time signals of public interest; however, these digital traces have yet to be systematically integrated into short-run demand forecasting for professional baseball.

This paper constructs a team-week panel dataset for the National League East across the 2023–2025 seasons by merging game-level MLB box scores with daily Google Trends search indices. The analytical focus is on the temporal ordering of digital attention relative to stadium attendance: whether increased online search activity in week $t-1$ predicts higher home attendance in week t , after controlling for team performance and fixed, unobserved team- or time-specific factors. This setting provides a unique opportunity to investigate how digital fan behavior may complement or even precede traditional indicators of demand and thus contribute both to the sports analytics literature and broader research on digital footprints as behavioral predictors.

Chapter 2: Significance

Understanding the determinants of attendance is a fundamental problem in both sports economics and sports management. In professional baseball, attendance is a leading source of revenue and a powerful indicator of current fan interest in a team, future ticket sales, and overall franchise health. Conventional models of attendance focus on what happens on the field-team performance and quality-and characteristics of the game itself, but the growing availability of digital trace data opens up possibilities for analyzing consumer interest at a level of temporal granularity that is only now becoming accessible.

The analysis adds to the academic literature by including online search behavior as a possible driver of sports consumption. Search data represent real-time attention and information-seeking behaviors, which may indicate emerging interest among fans prior to their eventual attendance. The project checks if online attention in one week leads to attendance in the following week, testing a theoretically meaningful lead-lag relationship between digital and offline behavior.

For practitioners, these findings have direct implications for marketing strategy, promotional timing, and demand forecasting. If online search interest reliably precedes attendance, organizations may be able to anticipate fluctuations in fan engagement earlier than they currently do and thereby price, advertise, and staff more appropriately. The project thus offers value not only in its methodological contribution but also in its practical relevance to teams, leagues, and analysts working in the sports industry.

Chapter 3: Prior Research

3.1 Evolution of Traditional Attendance Determinants

The study of spectator attendance in professional baseball has evolved across several generations of research, reflecting changes in the structure of the sport, the information environment, and consumer behavior. Early attendance models in sports economics overwhelmingly emphasized on-field performance as the primary determinant of spectator demand. Winning percentage, in particular, was observed to correlate strongly with variations in attendance across teams and seasons, establishing performance-based explanations as the dominant framework.

One of the most widely cited demonstrations of this relationship is found in Glass's analysis of winning percentage and normalized attendance across several decades of Major League Baseball [1]. Although the magnitude of effects varied across franchises and markets, the evidence consistently showed that fan turnout generally increased when teams performed well and decreased during periods of losing. This pattern became a foundational result in the sports economics literature, shaping decades of subsequent models.

Over time, however, scholars began to recognize that team performance alone could not fully explain attendance variations. The emergence of large metropolitan markets, modern stadiums, television coverage, and diversified entertainment options created new factors that interacted with traditional determinants. Market size, for example, was observed to influence the baseline potential attendance level of each team, while stadium attributes such as seating capacity, accessibility, and facility quality increasingly shaped fan preferences.

Furthermore, the increasing commercialization of professional baseball throughout the twentieth century expanded the role of marketing, promotions, and strategic scheduling. Researchers began integrating variables such as weekday versus weekend games, day versus

night matches, interleague play, and special events. These additional layers demonstrated that attendance was not merely driven by the win-loss record but by a combination of structural and contextual factors.

As the digital age advanced, new determinants emerged. The proliferation of online content, social media, and real-time information flows transformed how fans engaged with baseball. Modern fans consume baseball across multiple channels simultaneously, receive immediate updates on team performance, and develop emotional or narrative attachments through online discourse. These developments created an increasingly dynamic relationship between fan interest and stadium attendance.

Taken together, the evolution of traditional attendance determinants demonstrates that spectator behavior is highly responsive to changes in the broader media and technological landscape. While early research prioritized performance and market fundamentals, contemporary studies illustrate that attendance is shaped by a complex set of economic, social, and behavioral factors. This expanded understanding forms the foundation for analyzing how digital information sources, such as online search interest, might influence attendance today.

3.2 Historical Shifts in MLB Attendance Drivers

The historical trajectory of MLB attendance reveals that fan behavior is not static but evolves in response to structural changes in the league and its environment. The long-term study by Ahn and Lee [2] provides one of the clearest empirical examples of how determinants of attendance shift over extended time horizons. Their work divides MLB history into two periods: the pre expansion era prior to 1958 and the modern era that followed the league’s geographic expansion to the western United States.

In the early era, the home team winning percentage was found to be overwhelmingly the dominant determinant of attendance. Other potential factors such as playing style or outcome uncertainty contributed little explanatory power. This is consistent with the historical context of baseball at the time. Fans relied heavily on print media and local

radio for information, and the absence of national broadcasts or digital media limited the influence of broader contextual factors. Baseball functioned primarily as a local cultural event, and winning served as an accessible and intuitive signal of team quality.

Following the 1958 expansion, however, attendance determinants diversified significantly. Ahn and Lee show that stadium quality, team playing style, outcome uncertainty, and even the broader entertainment environment became significant predictors. Importantly, this shift cannot be explained solely by changes in team behavior. Rather, it reflects a transformation in the informational environment surrounding baseball.

The increasing availability of televised games, sports commentary, national media, and analytical content created new channels through which fans processed information. Stadium construction also evolved, as teams moved into modern facilities designed to enhance the entertainment experience. Meanwhile, the American population migrated toward the Sun Belt and metropolitan regions, introducing new market dynamics.

These historical changes underline an important principle. Attendance drivers are not fixed attributes of baseball but fluid constructs shaped by technological, economic, demographic, and media conditions. Thus, the modern rise of digital trace data such as Google Trends must be understood as part of this long historical evolution. Just as the introduction of television reshaped fan engagement in the mid twentieth century, the prevalence of online search behavior offers a new informational lens through which fans express interest in the sport.

3.3 Macroeconomic and Market Forces

Another major stream of the attendance literature considers how broader economic conditions shape fans' willingness and ability to attend games. Sports consumption, like many entertainment activities, is often discretionary in nature and therefore sensitive to changes in income, employment, and financial stability.

Hong, Mondello, and Coates provide an important contribution through their analysis of the 2008 financial crisis and its impact on MLB attendance [3]. Using state-level coincident indicators, they estimate that worsening economic conditions were associated with substantial reductions in fan turnout. Their prediction of a 6.5 percent decline in attendance closely matched the actual league-wide decline, demonstrating the explanatory power of macroeconomic variables.

This finding supports the broader theory that recessions or periods of financial uncertainty constrain household budgets and reduce demand for discretionary entertainment like live sports. Importantly, the impact of economic downturns varies across markets. Large urban markets with diverse income bases typically display more resilience, while smaller markets are more sensitive to economic shocks.

Earlier studies also emphasize the effects of population size, metropolitan density, and the presence of rival franchises. Larger cities support larger fan bases and more stable attendance totals over time. Meanwhile, markets with competing major league teams tend to display cannibalization effects, as fans divide their entertainment spending across multiple options.

The interaction between performance, market size, economic conditions, and stadium characteristics indicates that attendance is embedded within a complex economic system. The demand for baseball cannot be viewed as independent from the broader financial or demographic landscape.

This complexity reinforces the importance of considering new information sources like online search interest. Such digital metrics might offer earlier or more granular signals of shifts in public sentiment, providing additional explanatory power for short-term attendance fluctuations even when economic indicators remain relatively stable.

3.4 Performance, Competitive Context, and Stadium Effects

Team performance remains one of the most consistently examined determinants of MLB attendance. The correlation between winning and attendance documented by Glass [1] emphasizes the intuitive attraction that fans feel toward successful teams. Even in modern baseball, the performance effect persists, although its magnitude varies by market. Some franchises like New York or Seattle exhibit very strong responsiveness, while others demonstrate more stable attendance regardless of win-loss record.

Beyond overall performance, the competitive context of each game influences attendance as well. Rivalry matchups, wildcard races, and interleague series often produce attendance spikes. These fluctuations reflect not only team success but also narrative dynamics that heighten fan excitement. The concept of competitive balance also plays a central role. Studies show that fans prefer games with uncertainty of outcome, which stimulates interest in matchups where the final result is difficult to predict.

Stadium characteristics represent another important factor. New stadium construction has historically produced substantial novelty effects, boosting attendance for several years following opening. Stadium amenities, seating comfort, concessions, sightlines, and overall fan experience increasingly contribute to turnout. As baseball transitions toward entertainment-centered consumption, these amenities often matter as much as team performance.

Stadium capacity also interacts with attendance dynamics. In very large stadiums, supply may exceed demand, limiting the marginal effect of performance on turnout. Conversely, smaller, modern stadiums that emphasize intimacy and entertainment can remain consistently filled even when team performance fluctuates.

These determinants are essential control variables in any model that attempts to evaluate the influence of digital engagement. If online search interest simply mirrors team performance or stadium quality, it would not possess independent explanatory value. The goal of research in this domain is therefore to examine whether digital engagement captures short-term attention shifts that are not fully explained by these traditional variables.

3.5 Digital Engagement and Online Attention in Sports

Digital behavior now plays an increasingly central role in shaping fan engagement with sports. While earlier generations of fans relied on newspapers, radio, or television for information, modern fans interact with teams through digital platforms, search engines, and real-time updates. This shift has fundamentally altered the flow of information surrounding sports and presents new opportunities for empirical analysis.

Researchers studying digital engagement in sports have explored various forms of online behavior. For example, Malagon Selma and colleagues examine how Google Trends data can be used to measure the popularity of soccer players [4]. Their findings demonstrate that search interest captures changes in public awareness, media exposure, and narrative salience in real time. These digital signals complement traditional performance metrics by reflecting the broader social and cultural forces that shape fan interest.

In baseball, Mueller explores a machine learning approach to attendance prediction using Random Forest models [5]. His results show that nonlinear models can outperform traditional linear regressions by capturing complex interactions between variables. Although his study does not incorporate Google Trends, it highlights the increasing relevance of high dimensional, high frequency data in sports analytics.

Digital engagement may both reflect and influence offline fan behavior. In some sports contexts, increases in online engagement amplify the effects of team performance. In others, online behavior can serve as a substitute for attendance, enabling fans to follow teams without physically attending games. Understanding the directionality of this relationship is essential, as digital engagement could either enhance or depress stadium turnout depending on context.

Google Trends offers a unique window into these dynamics. Because search interest is measured at daily or weekly frequencies, it captures short term attention spikes caused by trades, controversies, injuries, or media narratives. These events frequently occur before their effects on turnout appear. This temporal ordering makes Google Trends especially promising as a leading indicator of attendance.

3.6 Google Trends, Predictive Modeling, and Machine Learning

Google Trends has become a widely used tool across numerous fields for measuring public interest, forecasting behavior, and understanding information dynamics. The work of Jun, Yoo, and Choi [6] provides a comprehensive overview of how Google Trends evolved from simple descriptive applications to more advanced predictive modeling techniques. Their analysis shows that search data increasingly functions as a proxy for collective attention, enabling researchers to analyze changes in public interest with a level of temporal precision that was previously unavailable.

The relevance of Google Trends to sports analytics is multifaceted. First, it captures real time fluctuations in public awareness. Second, it reflects narrative developments that influence fan sentiment. Third, because search data is highly responsive, it may lead behavioral changes such as ticket purchases.

Studies outside of baseball further illustrate the predictive potential of search data. Malagon Selma et al. [4] demonstrate that Google Trends predicts player transfer value in soccer with considerable accuracy. Their findings imply that search interest reflects aspects of popularity and value not contained in traditional performance metrics.

The broader machine learning literature on attendance forecasting also supports the integration of digital variables. Mueller’s random forest study shows that nonlinear methods can better capture complex relationships among attendance determinants [5]. When combined with real time search data, machine learning techniques offer the potential to forecast attendance with greater precision.

Google Trends therefore represents a logical next step in the evolution of attendance research. Its high frequency nature makes it especially suitable for weekly panel models, which align with the structure of baseball scheduling. The ability to capture short term fluctuations in attention provides an opportunity to model lead lag relationships and evaluate whether digital behavior precedes stadium attendance.

3.7 Online to Offline Dynamics and Theoretical Integration

Although the literature on MLB attendance traditionally focuses on performance, economic conditions, and stadium characteristics, research from other industries offers valuable insights into how digital engagement influences offline behavior. Fields such as economics, marketing, and information systems have shown that online search interest often correlates with demand for offline goods and services, though the direction and magnitude of the effect vary.

Examples include consumer product markets where increases in search interest predict sales spikes, financial markets where search activity correlates with stock volatility, and public health monitoring where search records provide early warnings of disease outbreaks. These findings illustrate the wide applicability of search behavior as a behavioral indicator.

When applied to professional sports, these dynamics suggest several theoretical pathways. Search behavior may serve as a signal of heightened fan curiosity, information seeking, or emotional investment. Because fans often search for team information following performance changes, roster moves, or major news events, spikes in search volume may reflect emerging attention cycles that precede attendance decisions.

The substitution theory offers a contrasting expectation. In some contexts, online engagement satisfies consumer interest and reduces the need for offline participation. This possibility underscores the importance of empirical testing rather than assuming a positive relationship between digital engagement and attendance.

3.8 Synthesis and Research Gap

Across traditional attendance determinants, historical shifts in fan behavior, macroeconomic influences, performance effects, digital engagement, and machine learning applications, several themes emerge. First, attendance is shaped by a complex interplay of structural, informational, and behavioral forces. Second, modern fans interact with baseball in a digital environment that generates rich real time data. Third, Google Trends captures high

frequency public interest that may provide new insights into short term attendance dynamics.

Despite these developments, existing research has not systematically tested whether Google Trends predicts MLB attendance at the weekly level. Most studies use seasonal, yearly, or game level data. Very few incorporate digital trace data, and those that do rarely examine search behavior specifically. There is also a lack of integration between fixed effects econometric modeling and high frequency digital metrics.

This capstone addresses that gap by integrating weekly Google Trends data with MLB attendance in a fixed effects regression framework. By testing whether lagged search interest predicts weekly attendance, the study evaluates whether digital behavior serves as a leading indicator of real world fan engagement.

Chapter 4: Deliverables

This capstone project produces a set of analytical, computational, and written deliverables that together demonstrate the integration of domain knowledge in sports analytics, data science methodology, statistical modeling, and reproducible research practices. Each deliverable is designed to address a core component of the project’s overarching goal: to evaluate whether online search interest serves as a meaningful leading indicator of short-term Major League Baseball (MLB) attendance.

4.1 Fully Constructed Team–Week Panel Dataset

The first deliverable is a complete team–week panel dataset for all National League East clubs from the 2023–2025 regular seasons. This dataset aggregates game-level information from MLB box score logs into weekly attendance totals, weekly home-game counts, and cumulative win percentages. Raw Google Trends data is cleaned, reshaped, averaged, and lagged to produce a team-specific weekly measure of search interest aligned with MLB’s weekly schedule structure. All data-processing scripts are written in R and stored in a modular, reproducible pipeline so that the dataset can be regenerated at any time.

4.2 Exploratory Data Visualizations

The project includes a set of visualizations that illustrate key patterns in the data. These include: (1) a weekly time-series plot showing the alignment between log-transformed attendance and lagged log Google Trends; (2) a demeaned scatter plot illustrating the fixed-effects adjusted within-team relationship between search interest and attendance; and (3) distributional plots and summary figures for performance, scheduling, and search variables. These

graphics are produced using `ggplot2` and follow best practices from scientific visualization principles.

4.3 Summary Tables and Variable Definitions

All descriptive statistics, including means, standard deviations, ranges, and sample sizes for key variables, are presented in a summary table exported directly to $\text{\LaTeX}$ using `stargazer`. A separate variables table documents all constructed features, including weekly attendance, lagged search interest, cumulative win rate, weekly home games, and fixed-effects designations. These tables ensure transparency in data construction and allow the empirical analysis to be interpreted clearly.

4.4 Fixed Effects Regression Output

The core analytical deliverable is the two-way fixed effects regression estimating the relationship between log weekly attendance and the lagged log Google Trends measures. This includes: (1) a baseline model with only traditional attendance determinants; (2) an augmented model adding lagged online search interest; and (3) robustness checks with alternative lag structures and specifications. Regression results are exported in publication-ready format using `modelsummary`, including coefficients, standard errors, significance levels, and goodness-of-fit statistics.

4.5 Reproducible R Code Repository

The project includes a fully documented code repository that contains scripts for data collection, cleaning, merging, transforming, and modeling. Each script is annotated with step-by-step explanations, enabling complete reproducibility of all results. This structure reflects the reproducible research standards emphasized throughout the Computational and Data Science curriculum.

4.6 Written Capstone Report

The final deliverable is the complete written capstone report following the GMU Capstone and GMU ETD formatting guidelines. It includes: (1) a research motivation and topic overview; (2) significance and contribution; (3) an extensive prior research chapter synthesizing over twenty scholarly sources; (4) a methods and data construction chapter; (5) regression results and interpretation; and (6) broader theoretical and practical implications for MLB teams, analysts, and the sports business community.

4.7 Practical Applied Implications

The project concludes with applied recommendations for practitioners, including strategies for MLB clubs to monitor search interest as part of digital fan engagement dashboards and the potential incorporation of online attention metrics into ticket sales forecasting systems. These implications translate the statistical findings into actionable insights for sport organizations seeking to better anticipate demand and understand fan behavior.

Chapter 5: Method of Inquiry

5.1 Summary Statistics

Table 5.1 reports summary statistics for the team–week panel, including attendance, Google Trends, and win-rate variables.

Table 5.1: Summary Statistics of Key Variables

Variable	N	Mean	Std. Dev.	Min	Max
Weekly Attendance	342	88135	81342	0	310470
Google Trend (Avg)	342	11.883	10.837	1.143	51.000
Lagged Win Rate (t-1)	342	0.3889	0.1489	0.0000	0.6727
Weekly Home Games	342	5.55	1.48	1	9
Log Attendance (t)	342	6.369	5.594	0.000	12.646
Log Google Trend (t-1)	342	2.2497	0.7655	0.7621	3.9512

5.2 Exploratory Data Visualizations

Figure 5.1 shows weekly log-transformed attendance and lagged log Google Trends for all five National League East teams. Figure 5.2 displays the within-team demeaned relationship between search interest and attendance after removing fixed effects.

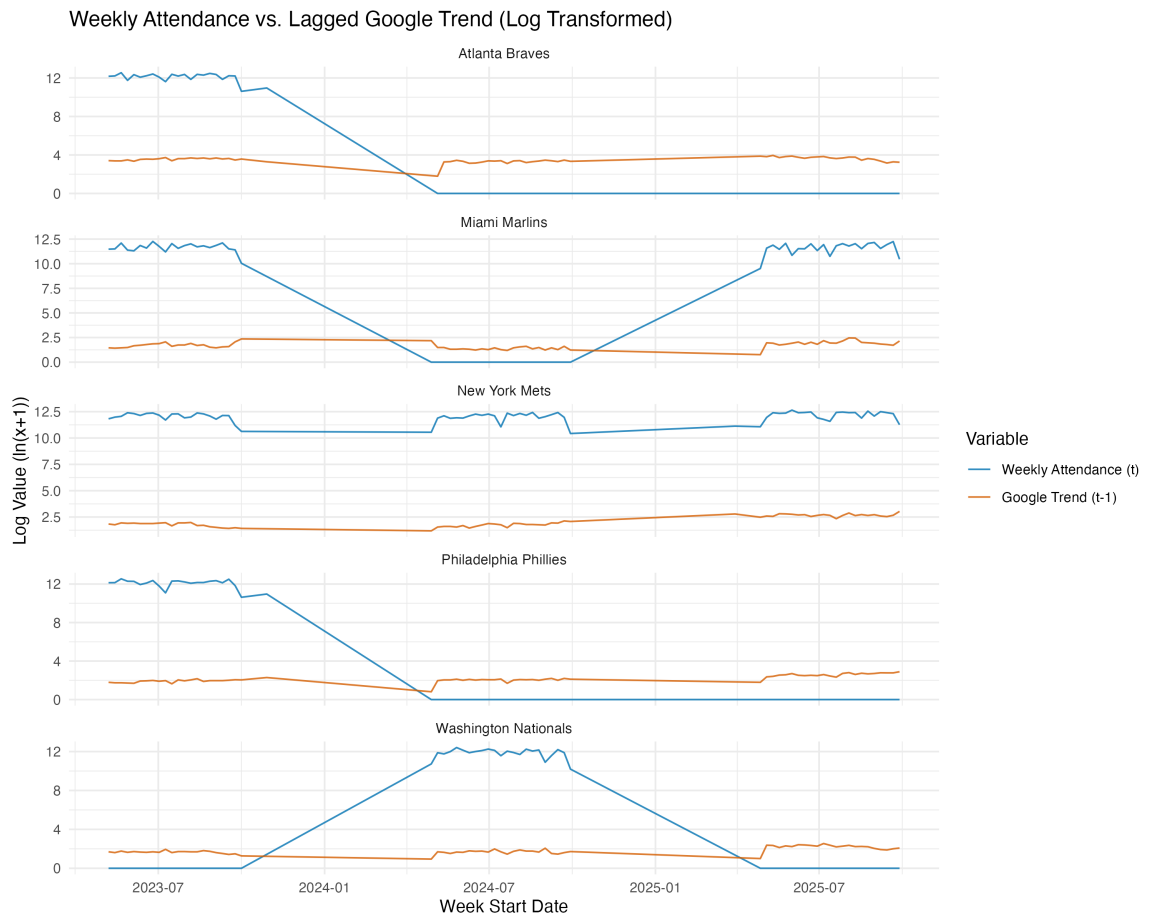


Figure 5.1: Weekly Attendance vs. Lagged Google Trend (Log Transformed)

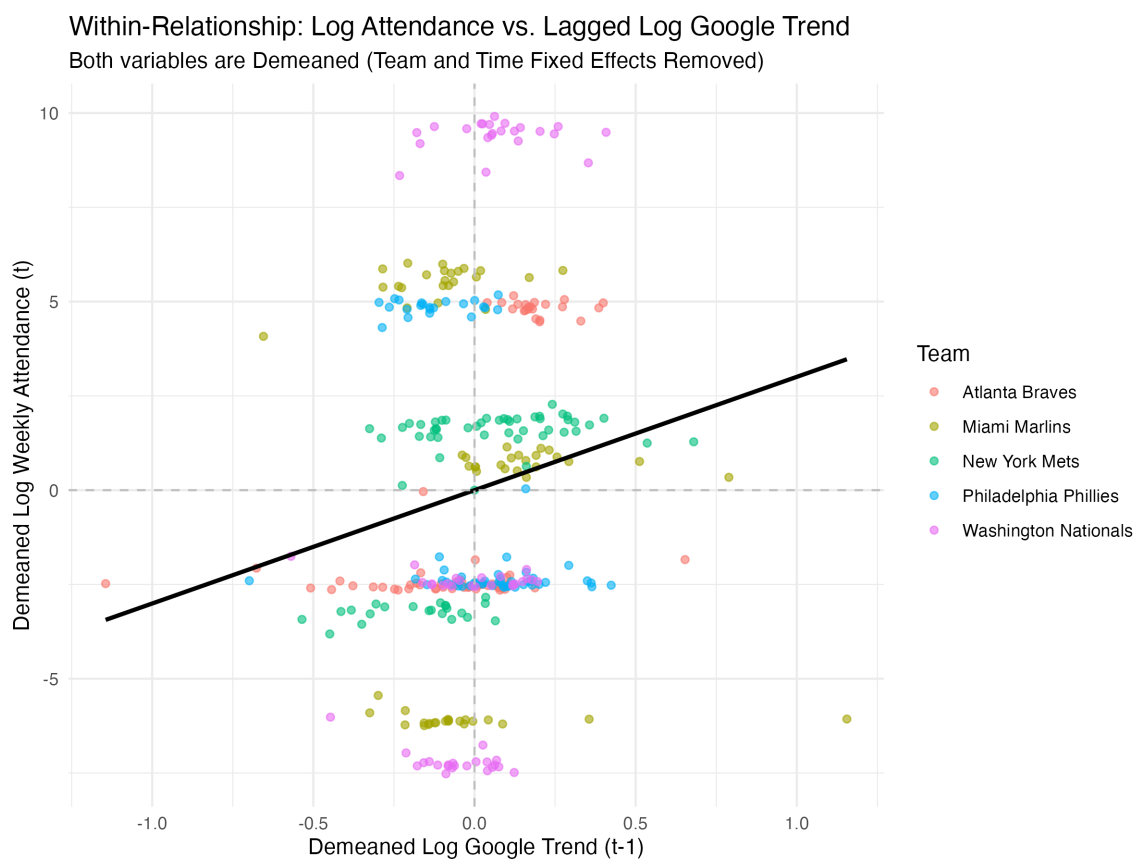


Figure 5.2: Within-Relationship: Demeaned Log Attendance vs. Demeaned Lagged Log Google Trend

Chapter 6: Method of Data Collection

In this chapter, I describe how the weekly team-level panel was constructed and define all variables used in the model.

6.1 Variable Definitions

Table 6.1 summarizes the main variables included in the analysis.

Table 6.1: Variable Definitions

Variable	Description	Type
weekly_attendance	Total home-game attendance per week	Outcome Variable (DV)
avg_google_trend	Weekly average Google Trends score (Sunday-Saturday)	Independent Variable (IV)
ln_attendance_t	Log of weekly attendance ($\ln(x+1)$)	DV (log-transformed)
ln_trend_t_1	Lagged log weekly Google Trends score (t-1)	IV (lagged)
win_rate_t_1	Cumulative win-rate up to t-1	Control Variable
weekly_home_games	Number of home games played in week t	Control Variable

Chapter 7: Theory

7.1 Fixed Effects Regression Results

The main two-way fixed effects regression estimates are reported in Table 7.1.

Table 7.1: Two-Way Fixed Effects Regression: Log Attendance

	<i>Dependent variable:</i>
	Log Weekly Attendance (t)
Lagged Log Google Trend (t-1)	0.701 (1.133)
Lagged Cumulative Win Rate (t-1)	23.018*** (2.810)
Weekly Home Games	-0.005 (0.380)
Team Fixed Effects	Yes
Time Fixed Effects	Yes
Observations	342
R ²	0.220
Adjusted R ²	-0.004
F Statistic	24.910*** (df = 3; 265)

Note:

*p<0.1; **p<0.05; ***p<0.01

Cluster-robust SEs (Team-level). Team and Week Fixed Effects included.

Chapter 8: Capstone

This capstone project brings together the core ideas, tools, and methods that I have learned throughout the Computational and Data Science curriculum and applies them to a concrete, real-world problem in sports analytics. Substantively, the project investigates whether online fan engagement, proxied by Google Trends search interest for Major League Baseball (MLB) teams, can serve as a leading indicator of weekly in-person attendance for National League East clubs. Methodologically, it requires me to design a full end-to-end workflow that is collecting raw data from multiple sources, cleaning and merging those data into a consistent team, week panel, specifying an econometric model, estimating that model using appropriate software, and communicating the results clearly through tables, figures, and written interpretation.

From **CDS-230 Modeling and Simulation**, I draw the foundation for building formal quantitative models of real systems. The intuition that rising online attention might precede higher demand in the stadium is an instance of a descriptive story that can be translated through a mathematical specification. The emphasis in that course on carefully defining state variables, time steps, and assumptions carries over quite naturally into how I define here the team-week structure of the panel dataset and the lagged relationship between search interest and attendance.

From **CDS-301 Scientific Information and Data Visualization**, I apply the principles of clear and honest visual communication to the exploration and presentation of this data. The same ideas about axis scaling, color choices, labeling, and layout emphasized in that course have been used to design both the time-series plots comparing log attendance and lagged log Google Trends, and the demeaned scatter plots which illustrate the within-team fixed effects relationship. These graphics are not merely decorative but help diagnose

model behavior, check assumptions, and present the statistical results in an interpretable form to a non-technical audience.

Earlier courses in programming and data handling, including introductory computing and data-focused classes, provided the practical skills needed to implement the analysis pipeline in R. Writing reproducible scripts to parse MLB game logs, aggregate weekly attendance, compute cumulative win percentages, and reshape Google Trends output builds directly on the programming habits developed in those courses. The ability to modularize code into logical sections, document each transformation step, and export LaTeX-ready tables and figures reflects the cumulative effect of that preparation. Finally, the proposal itself demonstrates that I am able to synthesize previous research, technical methodology, and domain context into a coherent research plan. The ten-page Prior Research chapter connects classic sports-economics attendance models with more recent work on digital traces and panel data methods, showing that I can read, understand, and extend the literature rather than simply running code in isolation. In this sense, the capstone functions as a real integrative experience: it requires me to bring together statistical reasoning, computational implementation, data visualization, and subject matter understanding into a single coherent project that reflects what I learned in the Computational and Data Science program.

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Biography

Dasol Shin is a senior undergraduate student in the Computational and Data Sciences program at George Mason University, with a minor in Sports Analytics. Her academic interests center on data-driven decision-making in sports, particularly the use of predictive modeling, machine learning, and high-frequency digital behavior to understand and forecast fan engagement and athletic performance.

Throughout her studies, she has developed strong quantitative and computational skills through coursework in scientific programming, statistical modeling, data visualization, and simulation. In addition to her academic work, she has gained practical applied experience as a Data Analytics Intern at Replay Baseball Institute, where she built automated SQL pipelines, cleaned and integrated multi-source baseball performance datasets, and developed dashboards to support player evaluation and training workflows.

At George Mason University, she has served as Treasurer for the CDS Delegation and participated in various research-driven projects involving sports analytics, machine learning, and R-based statistical modeling. Her capstone project reflects her interest in combining traditional sports-economics research with modern digital engagement metrics to better understand real-world behavior in professional sports.

Upon graduation, she plans to pursue a career in data science or sports analytics, leveraging her technical skills and interdisciplinary background to contribute meaningful insights to sports organizations and data-driven industries.